COMMUNICATION

Paleomagnetic dating of the transformation of oolitic goethite to hematite in iron ore

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We show that the oolitic hematite ores of Birmingham, Alabama, carry a remanence direction stable enough to have survived major folding. Nevertheless, this remanence was very likely acquired in the Pennsylvanian, about 130 Ma after the ore's Early Silurian deposition. This long delay is understandable if we accept the common hypothesis that the ore originated as oolitic goethite. We would then not expect acquisition of a stable remanence until deep burial in the Pennsylvanian raised the temperature to the ~80°C probably needed to transform coarse goethite to hematite in water. Although the paleomagnetism of oolitic hematite may be of limited value in defining primary paleopoles, it may allow dating of when the ore's temperature was first raised to ~80°C, perhaps aiding studies of sedimentary basin evolution and oil formation. Our results also support using caution in interpreting the paleomagnetism of more common red beds, emphasizing that a positive fold test is not necessarily evidence of primary remanence and that burial history may control the timing of remanence acquisition.

Nous montrons que l'hématite oolithique des minerais de Birmingham, Alabama, porte une direction de rémanence suffisamment stable pour avoir servécu à une phase importante de plissement. Néanmoins, cette rémanence fut vraisemblablement acquise au Pennsylvanien—soit environ 130 Ma après la déposition des minerais au Silurien inférieur. Ce long retard est compréhensible si on accepte l'hypothèse généralement admise que le minerai doit son origine à la transformation de la goethite oolithique. L'acquisition de la rémanence stable n'a pu se produire qu'après un enfouissement profond au Pennsylvanien accompagné d'une élévation de température jusqu'à environ 80°C, possiblement nécessaire pour assurer la transformation de la goethite à grain grossier en hématite dans de l'eau. Bien que le paléomagnétisme de l'hématite oolithique soit d'application limitée pour définir les paléopôles primaires, il peut cependant permettre de dater la période à laquelle la température a atteint pour la première fois environ 80°C, aidant probablement à mieux comprendre les études de l'évolution d'un bassin sédimentaire et de la formation d'huile. Nos résultats incitent à la prudence quant à l'interprétation du paléomagnétisme des couches rouges plus fréquentes et démontrent que le test positif de plissement n'indique pas nécessairement une rémanence primaire et que l'histoire de l'enfouissement peut contrôler la période d'acquisition de la rémanence.

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Oolitic hematite, the most common sedimentary iron ore of Paleozoic age, often carries a single stable direction of remanent magnetization (Buchan and Hodych 1982*a*,*b*; Howell *et al.* 1958; Andreeva *et al.* 1965). However, this remanence may often be acquired long after deposition as we show in a paleomagnetic study of oolitic hematite-bearing sediments of the Silurian Red Mountain Formation of Alabama. Oriented samples, many of probable ore grade, were collected at sites on both limbs of the Birmingham Anticlinorium (Fig. 1). The natural remanent magnetization (NRM) directions are well grouped at most sites, as can be seen in Table 1 from α_{95}, a measure of angular scatter about the site mean remanence direction. (This agrees with work by Howell *et al.* (1958) on the NRM of ore at one site near Birmingham.) Each sample was demagnetized by heating to successively higher temperatures and cooling in zero field, usually using temperatures of 200, 300, 400, 500, 550, 600, 640, 660, and 670°C. Demagnetizing through 600 or 640°C caused little change in remanence direction and either reduced scatter at a site or caused little change in scatter (Table 1). Intensity of remanence decreased slowly on demagnetizing to 600 or 640°C (see M/M_i in Table 1) before falling rapidly near the Curie temperature of hematite. Thus, the samples have a single stable direction of remanence mostly carried by hematite with high blocking temperatures (600–670°C).

When was the stable remanence direction of the Red Mountain Formation acquired? It was acquired later than the time of deposition since cobbles of oolitic hematite-bearing limestone (Bearce 1973) in an intraformational conglomerate (site 1c) have the same *in situ* remanence direction as the ores above (sites 1a and 1b) despite great variation in tilt of internal bedding among the 11 cobbles measured. However, stable remanence was acquired before the Red Mountain Formation was folded (Buchan and Hodych 1982*a*) since the two limbs of the Birmingham Anticlinorium have very different remanence directions before correction for tilt of bedding, but not after (Fig. 2). Folding probably occurred in the Pennsylvanian, since sediments of Early Pennsylvanian age are included in the folding (King 1950), and yet red beds farther north in the Valley and Ridge Province show probable Pennsylvanian post-folding remagnetization (Van der Voo 1979). The paleopole yielded by the remanence direction of the Red Mountain Formation is in good agreement with the well defined mean Pennsylvanian paleopoles for North America (Irving and Irving 1982) but differs from Silurian (Roy *et al.* 1967; French and Van der Voo 1979), Devonian, and Mississippian (Irving 1979) paleopoles (Fig. 3). Furthermore, all the samples have south-directed remanence, which is characteristic of most of the Pennsylvanian (Irving and Pullaiah 1976). Hence, the remanence of the Red Mountain Formation was very likely acquired in the Penn-

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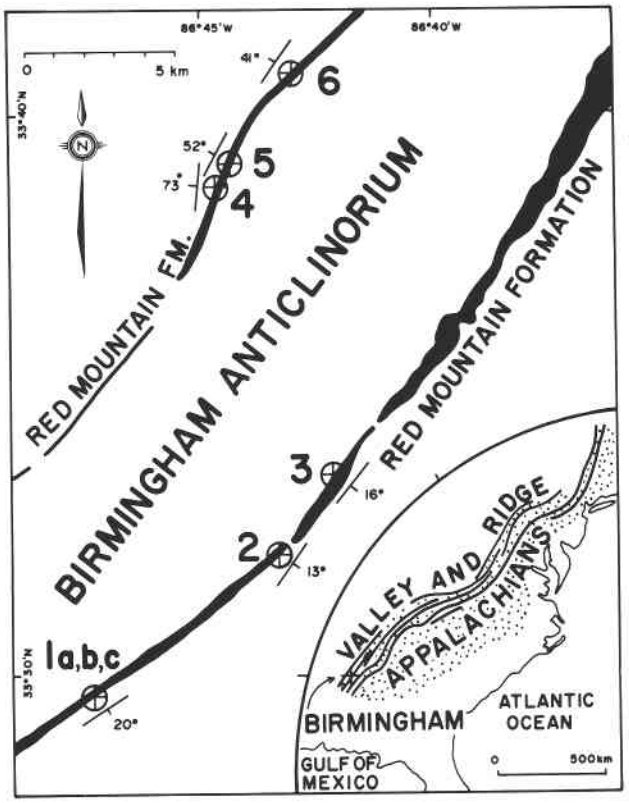


FIG. 1. Paleomagnetic sampling sites (numbered) in the Silurian Red Mountain Formation near Birmingham, Alabama (after Burchard and Butts 1910). The sites lie on both limbs of the Birmingham Anticlinorium located in the Valley and Ridge Province of the southern Appalachian Mountains shown in the inset. The sediments sampled are sandy (except at site 1c) and all bear oolitic hematite (of possible ore grade except at sites 3 and 5). Two additional sites just off the map area carry unstable remanence and are not used.

sylvanian ~130 Ma after deposition but shortly before folding. (Further support for a pre-folding Carboniferous acquisition of remanence in the Red Mountain Formation is provided in an abstract by Van der Voo *et al.* (1983), which came to our attention while this paper was being written.)

The long delay in acquisition of a stable single-component remanence in the Red Mountain Formation is most easily explained if we accept that most oolitic iron ores originated as goethite, which transformed to hematite much later. This popular hypothesis (James 1966) does provide a good uniformitarian explanation for why geothite is much more common than hematite in oolitic iron ores deposited in the last 180 Ma whereas hematite becomes much more common in older oolitic ores. Transformation of oolitic goethite to hematite should produce a chemical remanent magnetization (CRM) that can be stable (Hedley 1968), and delay of transformation for even more than 130 Ma is common (James 1966). In contrast, if the ore had originated as hematite, a Silurian remanence component would be expected (unless the hematite were too fine grained during the Silurian, but then long-continuing grain growth and complex multicomponent remanence would be expected). Certainly, the remanence cannot be a detrital remanent magnetization because it was not acquired during deposition. Nor can it be a thermal remanent magnetization because the blocking temperatures of the hematite (>600°C) are mostly too high for significant remagnetization (Pullaiah *et al.* 1975) at the maximum temperature experienced by the Red Mountain Formation (<200°C, judging by the colour of conodonts in nearby Ordovician limestones (Harris *et al.* 1978)).

Why the Red Mountain Formation finally acquired stable remanence in the Pennsylvanian before folding is understandable if we accept that transformation of oolitic geothite to hematite in the subsurface commonly requires heating to ~80°C. In theory (Langmuir 1971), fine-grained goethite (<0.1 μm) should spontaneously transform, whereas coarse-grained goethite (>1 μm) should require heating to ~80°C to transform to hematite in the high humidity expected in the subsurface. Although fine crystals (<0.1 μm) seem typical of oolitic goethite (Rohrlich 1974), they are presumably packed tightly enough into the ~1 μm flakes visible under the scanning electron microscope (Rohrlich 1974) to behave as coarse grains. Otherwise it is difficult to explain the common persistence of oolitic goethite for up to 180 Ma (James 1966). Assuming a normal geothermal gradient of 26°C/km (Brooks 1981), burial under ~2.3 km of sediment is required to attain the ~80°C probably needed to transform oolitic goethite pro-

TABLE 1. Remanent magnetization of oolitic hematite-bearing sediments of the Red Mountain Formation corrected for tilt of bedding

	Site mean remanence directions							
	Before demagnetization			After thermal demagnetization*				
Site	D	I	α_{95}	D	I	α_{95}	M/M_i	N
1a	150.2	23.5	2.7	150.2	20.3	2.5	0.66	5
1b	146.5	20.8	4.5	147.1	15.5	3.7	0.54	5
1c†	151.2	21.4	3.1	150.6	17.1	3.7	0.60	11
2	155.7	21.0	6.9	155.0	15.0	10.5	0.62	4
3	151.9	31.0	15.8	149.7	26.5	9.6	0.61	6
4	141.5	21.4	5.9	139.1	20.3	5.6	0.66	7
5	153.7	27.8	14.3	152.0	25.7	15.0	0.53	5
6	157.1	22.4	6.0	154.4	20.1	6.3	0.72	8

NOTES: D is the mean declination; I is the mean inclination; α_{95} is the half angle of the cone of 95% confidence about the mean; M/M_i is the fraction of remanence remaining after demagnetization; N is the number of samples.

*Demagnetized to 640°C, except at sites 1c and 6, which were demagnetized to 600°C.

†Eleven cobbles of intraformational conglomerate corrected for dip of conglomerate bed rather than dip of internal bedding of individual cobbles.

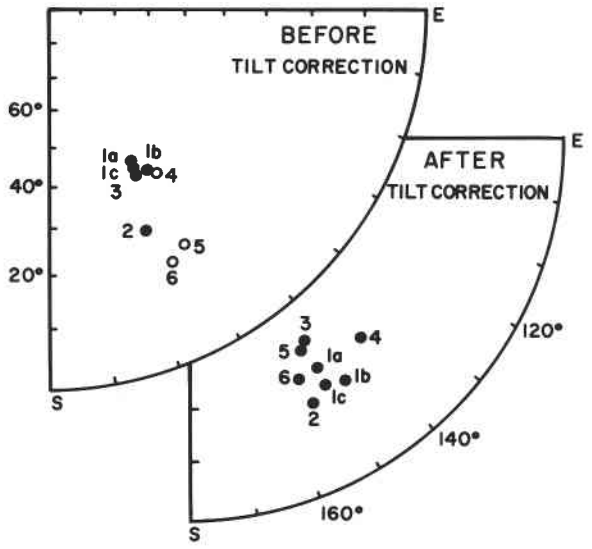


FIG. 2. Mean remanence directions of sites in the Red Mountain Formation (after thermal demagnetization as in Table 1) shown in stereographic projection. Before correcting for tilt of bedding, the remanence directions point downwards (closed circles) on the southeast limb of the Birmingham Anticlinorium but upwards on the northwest limb (open circles). After tilt correction, remanence directions on the two limbs do not differ significantly, showing that remanence was acquired before folding.

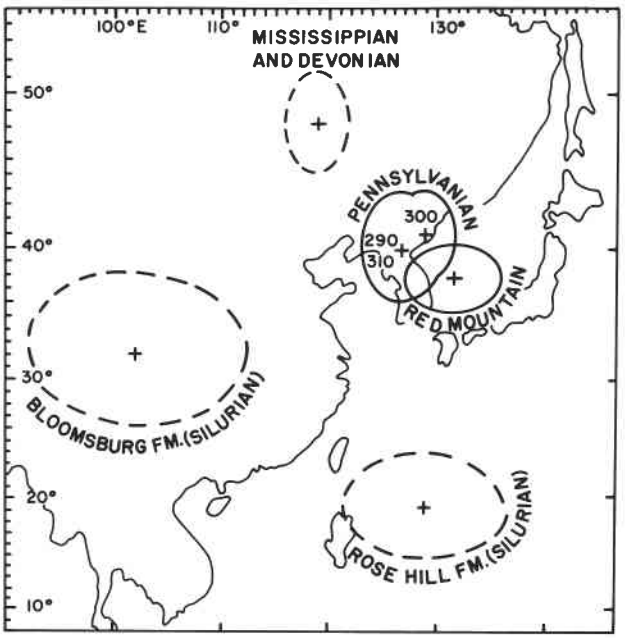


FIG. 3. The Red Mountain Formation paleopole shown at 37.6°N and 131.9°E with its oval of 95% confidence having semi-major and semi-minor axes dm = 4.6° and dp = 2.4° (giving unit weight to each of the thermally demagnetized site mean remanence directions of Table 1). This paleopole is in good agreement with the mean 290, 300, and 310 Ma old Pennsylvanian paleopoles for North America (Irving and Irving 1982). Also shown are Silurian paleopoles for two North American sedimentary formations (Roy *et al.* 1967; French and Van der Voo 1979) and the mean of Devonian and Mississippian paleopoles for stable North America (Irving 1979).

ducing CRM. There is no evidence for this deep a burial in the Birmingham area until the Pennsylvanian since the combined thickness of the overlying Devonian and Mississippian sediments is only ~1 km (King 1950). However, at least 1.5 km of Pennsylvanian sediments (King 1950) was added before folding, giving deep enough burial to produce the pre-folding Pennsylvanian remanence of the Red Mountain Formation.

If oolitic hematites have mostly transformed from goethites at ~80°C, they will be of limited value for establishing primary paleopoles unless they were quickly and deeply buried. For example, an attempt (Buchan and Hodych 1982*b*) to establish an Ordovician paleopole for the Avalonian microcontinent using the stable remanence of Ordovician oolitic hematite ores may have yielded a Devonian paleopole since the ores may never have been deeply buried but were in a region of Devonian low-grade metamorphism. Although oolitic hematite is not normally present in continental red beds, other colloform hematites have been reported (Turner and Ixer 1977); these may also have dehydrated from goethite at ~80°C, adding to the complexity of remanence often reported (Turner 1980) in continental red beds. Our results emphasize that the paleomagnetism of red beds should be interpreted with caution. A positive fold test is not necessarily evidence of primary remanence and burial history may control the timing of remanence acquisition through diagenetic chemical changes.

A test of whether oolitic hematites have commonly transformed from goethites at ~80°C may be possible by estimating the maximum temperature experienced by many oolitic hematites and goethites using indicators such as conodont colour and vitrinite reflectance. If the test is positive, paleomagnetism may allow dating of when oolitic hematites were first heated to ~80°C, perhaps aiding studies of the history of heating and oil maturation in sedimentary basins.

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